Dataset Assessment Document (DAD)

Logistics Analytics Dashboard | Dataset Assessment

Dataset Overview

Item	Details
Dataset Title	Logistics Orders Dataset
Domain / Theme	Logistics, supply chain operations, delivery performance, revenue and cost analytics
Source	Internship project files: logistics_unclean.xlsx and Logistic Clean.xlsx
File Type	Excel workbook (.xlsx)
No. of Rows	Raw dataset: 5,000 rows; cleaned dataset: 4,905 rows
No. of Columns	20 columns

Data Structure Summary

Column Name	Data Type	Dimension / Measure	Description
Order_ID	String	Dimension	Unique order identifier used to count and track logistics orders.
Customer_ID	String	Dimension	Customer reference ID for order-level customer analysis.
Order_Date	Date	Dimension	Date when the order was placed; supports year, month, quarter, and trend analysis.
Delivery_Date	Date	Dimension	Actual or expected delivery date used with order date to evaluate delivery timelines.
Carrier	String	Dimension	Shipping carrier name used for carrier performance and cost comparison.
Status	String	Dimension	Current shipment status such as Delivered, Pending, Returned, Cancelled, or In Transit.
Category	String	Dimension	Product category used for revenue, discount, weight, and shipment analysis.
Region	String	Dimension	Delivery region used for geographic and operational performance comparison.
Country	String	Dimension	Destination country used for map and country-level analysis.
Warehouse_ID	String	Dimension	Origin warehouse identifier used for warehouse workload and capacity analysis.
Driver_ID	String	Dimension	Assigned driver identifier used for driver workload

			analysis.
Weight_kg	Float	Measure	Package weight in kilograms used for shipment weight and warehouse load analysis.
Quantity	Integer	Measure	Number of units ordered or shipped.
Unit_Price	Float	Measure	Price per unit in USD used to calculate gross revenue.
Shipping_Cost	Float	Measure	Shipping fee in USD used to compare logistics cost by carrier and time.
Discount	Float	Measure	Discount rate applied to the order, used to calculate net revenue and discount impact.
Delivery_Days	Integer	Measure	Number of days taken for delivery, used for on-time and late-delivery analysis.
Payment_Mode	String	Dimension	Payment method used for the order.
Priority	String	Dimension	Shipment priority level used for priority-based operational analysis.
Customer_Rating	Float	Measure	Customer satisfaction rating, expected on a 1 to 5 scale.

Data Quality Assessment

Aspect Checked	Observations	Action Needed
Missing Data	Raw data had missing values in Carrier (147), Region (163), Warehouse_ID (138), Driver_ID (131), Shipping_Cost (171), Discount (148), Payment_Mode (132), and Customer_Rating (141). Cleaned data resolved most dimension nulls but retained missing Shipping_Cost (167), Customer_Rating (139), Unit_Price (7), and Weight_kg (2).	Replace key dimension nulls with controlled unknown labels; review remaining measure nulls before final KPI calculations.
Duplicates	Raw data contained 95 duplicate Order_ID records. Cleaned dataset has 0 duplicate Order_ID records.	Use cleaned dataset for dashboard calculations and keep Order_ID uniqueness as a validation check.
Outliers	Customer_Rating had documented out-of-range values such as 0, 6, 7, and 99 in the raw data. Quantity had documented negative values. Unit_Price had extra digit/string issues.	Validate business ranges: ratings 1-5, quantity greater than 0, discount between 0 and 0.40, and positive price/cost/weight values.
Incorrect Data Types	Raw data included dates stored in non-standard string formats and numeric values stored as strings with symbols or suffixes.	Convert Order_Date and Delivery_Date to date type; convert Weight_kg, Unit_Price, Shipping_Cost, Discount, Delivery_Days, Quantity, and Customer_Rating to numeric types.
Consistency Issues	Raw data had casing inconsistencies across Carrier, Status, Category, Region, Country, Payment_Mode, and Priority.	Standardize text casing and category labels before modeling and visualization.

Key Metrics & KPIs Identified (Initial)

KPI	Why It Matters
Total Orders	Shows logistics activity volume and supports trend, warehouse, driver, and carrier workload analysis.
Gross Revenue	Measures total business value before discounts and supports category, region, and country revenue analysis.
Net Revenue	Shows realized revenue after discount impact and supports finance-focused performance review.
On-Time Rate %	Indicates delivery reliability and service quality.
Late Delivery Rate %	Highlights operational delay risk and helps identify carriers/regions needing improvement.
Average Delivery Days	Shows delivery speed and supports carrier and regional performance comparison.
Average Customer Rating	Represents customer satisfaction and helps connect delivery performance to experience.
Shipping Cost	Supports cost control and carrier cost benchmarking.
Average Weight Per Order	Helps assess warehouse and transportation load.
Driver Workload	Identifies driver utilization and potential workload imbalance.

Assumptions / Risks

- The cleaned dataset will be used as the primary source for Tableau dashboard calculations, while the raw dataset is used to document data quality issues.
- Missing measure values such as Shipping_Cost and Customer_Rating may affect cost and satisfaction KPIs if not handled consistently.
- Unknown Carrier, Unknown Region, and WH-UNKNOWN values should be retained as explicit categories rather than hidden, so data gaps remain visible to business users.
- On-time and late-delivery calculations require a clearly defined delivery-day threshold; the dashboard currently treats late deliveries as a key delivery performance measure.
- Revenue calculations assume gross revenue is derived from Quantity multiplied by Unit_Price and net revenue is derived after applying Discount.
- Customer ratings are assumed to be valid only within the 1 to 5 range after cleaning.